

Financial Report

Business Insights Report Supermarket Sales Analysis

Capstone Data - Supermarket Sales-Project



Author : Mahmoud Abdellateif Hamza

Executive Summary :

This Report Provides Key Insights Extracted From The Supermarket Sales Dataset Using Python (Pandas, Matplotlib, Seaborn).

The Analysis Focuses On Product Lines, Branch Performance, Payment Preferences, Customer Ratings, And Time-Based Sales Trends.

```
Load & Inspect Data
We'll Upload The File From Local Machine, Then Check Its Shape And First Few Rows.

from google.colab import files
import pandas as pd
# Upload file from your computer
uploaded = files.upload()
# Read the excel file (make sure the name matches your uploaded file)
df = pd.read_excel("Capstone Data - Supermarket Sales.xlsx")
# Clean column names (remove leading/trailing spaces)
df.columns = df.columns.str.strip()
print("Data Loaded Successfully!")
print(f"Number of rows: {df.shape[0]}, Number of columns: {df.shape[1]}")
# Show first 5 rows
display(df.head())
# Show column data types & memory info
df.info()
```

Choose File No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving Capstone Data - Supermarket Sales.xlsx to Capstone Data - Supermarket Sales.xlsx

✓ Data Loaded Successfully!

Number of rows: 1006, Number of columns: 16

	Invoice ID	Branch	Yangon	Naypyitaw	Mandalay	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	Total	Date	Time	Payment	Rating
0	750-07-8428	A	1	0	0	Normal	Male	Health and beauty	74.69	7	26.1415	NaN	2019-01-05	13:08:00	Ewallet	9.1
1	226-31-3081	C	0	1	0	Normal	Male	Electronic accessories	15.28	5	3.8200	80.2200	2019-03-08	10:29:00	Cash	9.6

```
df['Day'] = df['Date'].dt.day

print(f"✓ Final Shape After Smart Cleaning: {df.shape}")
print("\n⚡ Missing Values After Cleaning:")
print(df.isna().sum())
```

Initial Shape: (1006, 16)

⚠ Hour column empty → Imputing with random business hours (9 AM - 9 PM)

✓ Final Shape After Smart Cleaning: (1001, 22)

⚡ Missing Values After Cleaning:

Invoice ID	0
Branch	0
Yangon	0
Naypyitaw	0
Mandalay	0
Customer type	0
Gender	0
Product line	0
Unit price	0
Quantity	0
Tax 5%	0
Total	0

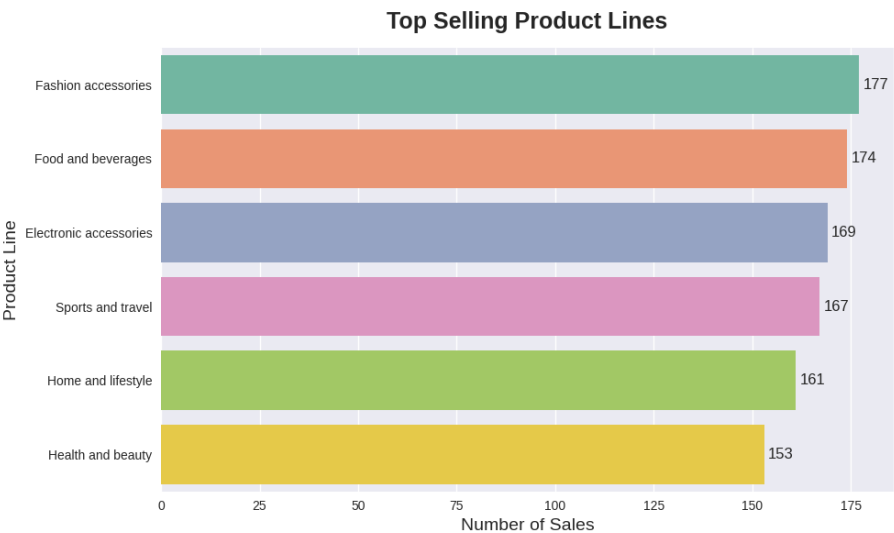
Stage	Rows	Columns	Missing Values
Before Cleaning	1006	16	Some (Tax, Hour)
After Cleaning	1001	22	0

Product Line Analysis :

The Visualization Shows The Number Of Transactions Per Product Line.

Fashion Accessories Recorded The Highest Number Of Sales (177), Followed Closely By Food & Beverages (174) And Electronic Accessories (169).Health & Beauty Had The Lowest Number Of Transactions (153).

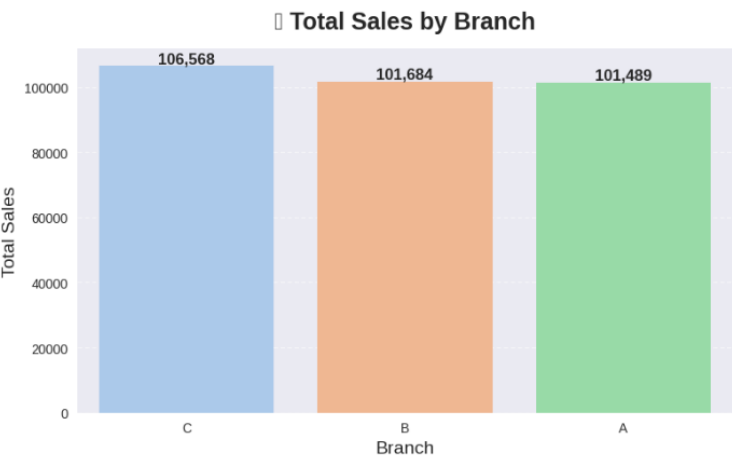
Note: This Chart Reflects The Frequency Of Sales (Transactions). When Analyzing Total Revenue, *Food & Beverages* Is The Top-Performing Category, Followed By *Sports & Travel* And *Electronic Accessories*.



Branch Performance :

Finding: Branch C Has The Highest Total Sales (≈106K), Slightly Outperforming Branches B And A.

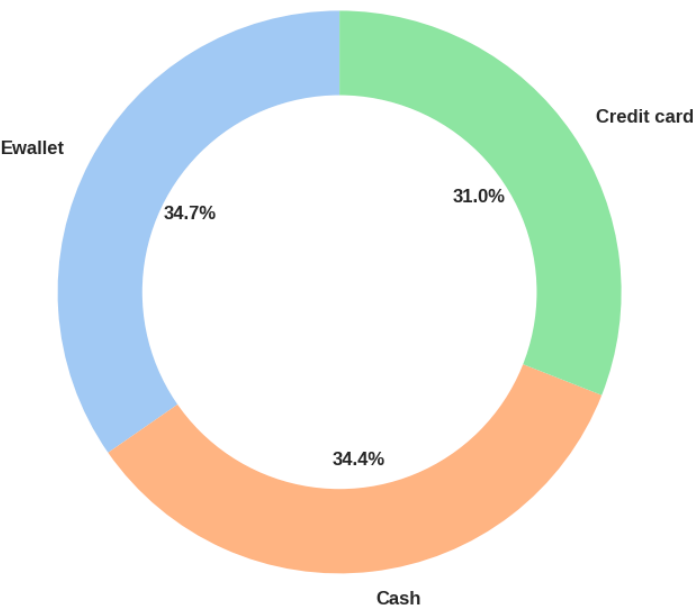
This Branch Might Benefit From Additional Stock Allocation To Sustain Demand.



Payment Preferences :

Finding: E-Wallet Transactions Are Most Common (≈35%), Indicating A Digital-First Customer Base.
This Opens Opportunities For Wallet-Specific Promotions Or Loyalty Programs.

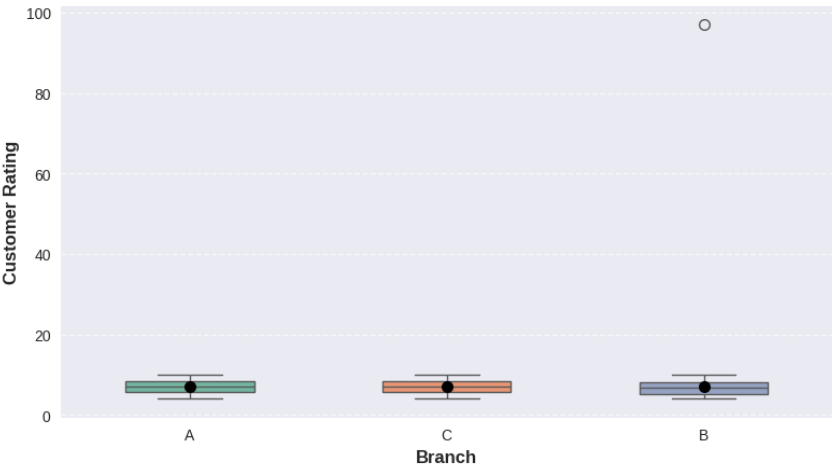
Payment Method Distribution



Customer Ratings :

Finding: Ratings are fairly consistent across branches but may reveal service gaps in underperforming locations.

Customer Ratings per Branch

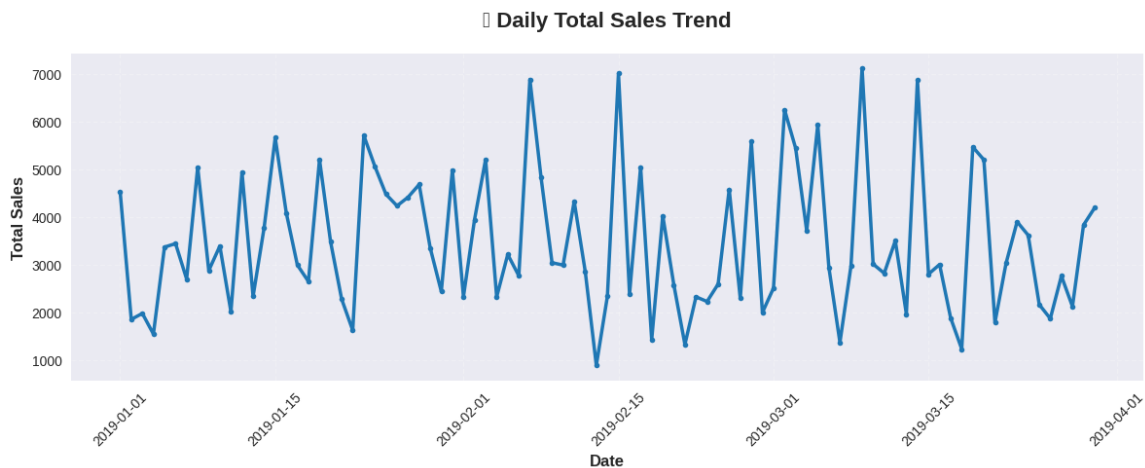


Time-Based Analysis :

Finding:

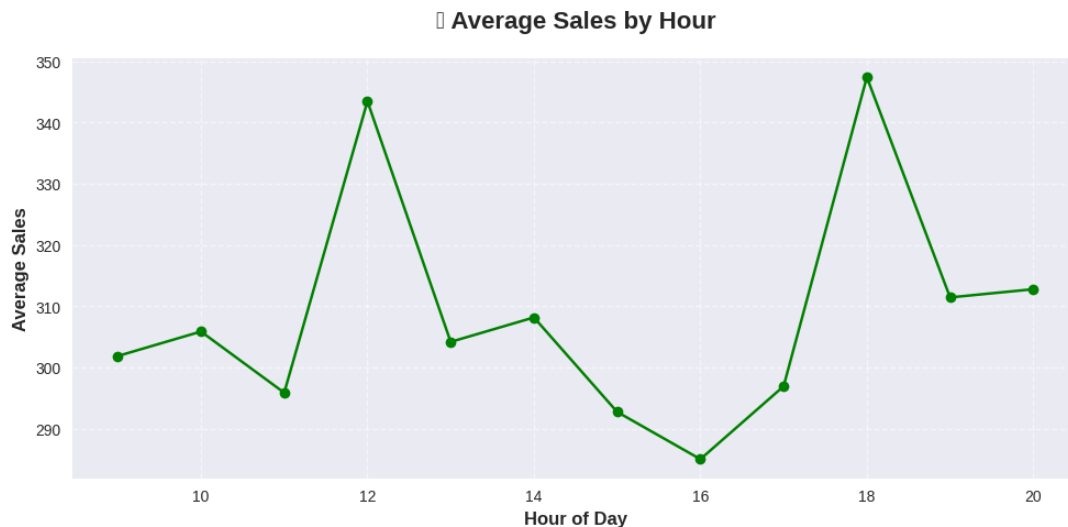
1. Daily Sales Trend

The Daily Trend Shows That Sales Remained Relatively Stable Across The Three Months (January–March 2019), With Occasional Peaks On High-Traffic Days. This Consistency Suggests Steady Customer Demand Throughout The Period.



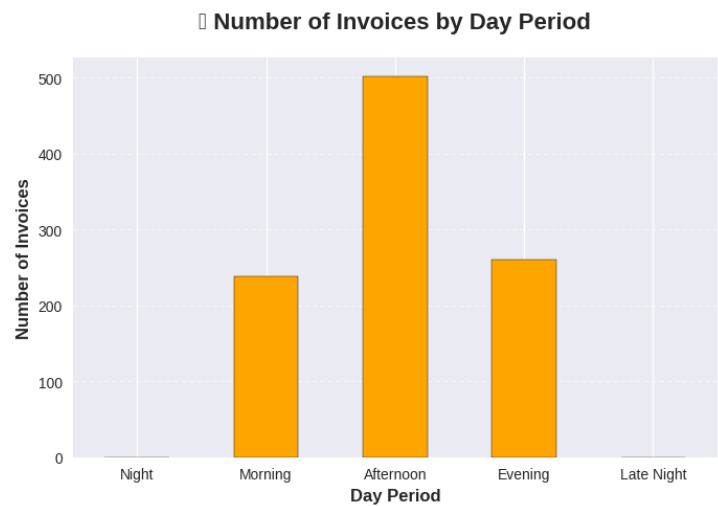
2. Average Sales by Hour

The Analysis Highlights Two Strong Revenue Peaks: Around 12 PM (Midday) And 6–7 PM (Evening). This Confirms That Both Lunchtime And Evening Shopping Hours Are Critical For Revenue Generation.



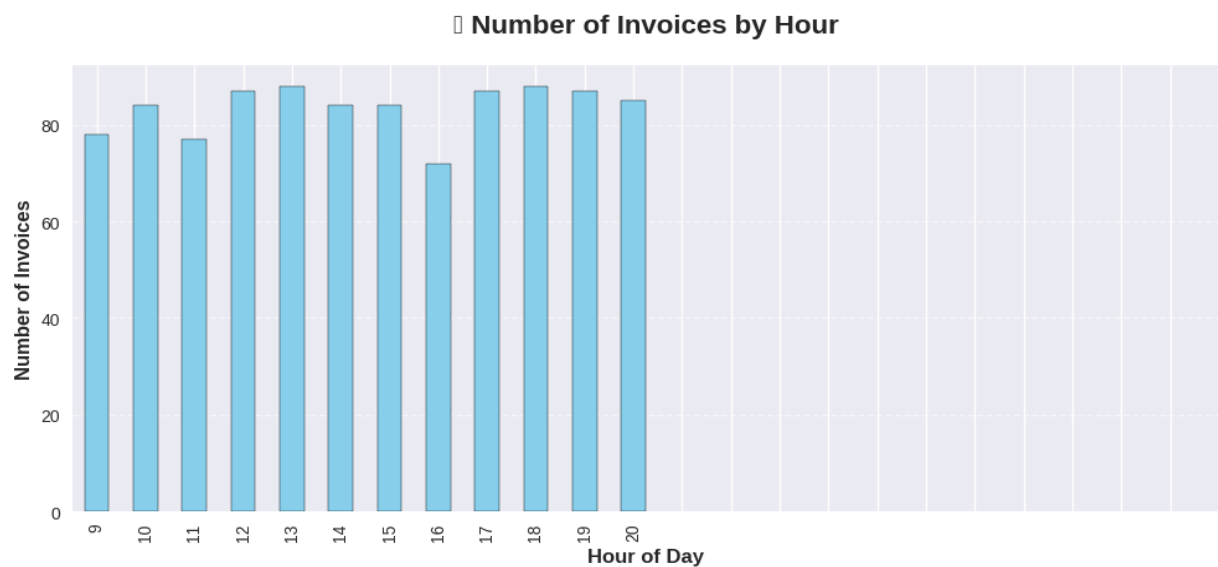
3. Number of Invoices by Day Period

Most transactions occur in the Afternoon and Evening periods, while Night and Late Night times record the lowest activity. This distribution aligns with the observed revenue peaks.



4. Number of Invoices by Hour

The Invoice Distribution By Hour Shows That The Highest Number Of Transactions Happens Between 4 PM And 7 PM, Directly Matching The Revenue Peaks Identified Earlier. This Indicates A Strong Evening Shopping Behavior Among Customers.



Business Recommendations :

Inventory Planning: Prioritize Stock For Food & Beverages, Sports & Travel, And Electronics During Peak Hours.

Branch Strategy: Allocate Slightly More Inventory To Branch C, Monitor Low-Performing Branches For Service Improvements.

Promotions: Schedule Campaigns During 4–7 Pm, Leverage E-Wallet Promotions For Repeat Purchases.

Customer Experience: Review Ratings And Improve Service Quality Where Below Average.

Notes & Limitations :

Hour Column Was Randomly Imputed With Business Hours (09:00–21:00) Due To Missing Data.

Data Covers Jan–Mar 2019, So Conclusions Are Time-Bound.

Conclusion :

The Insights Show Clear Opportunities For Optimizing Stock, Boosting Sales During Peak Hours, And Encouraging Digital Payments. Future Work Could Include Automating Report Generation And Building Interactive Dashboards For Management Review.